

Python Fundamentals - 57 Interview Questions with Answers

1. What is Python, and why is it considered a popular programming language?

Python is a high-level, interpreted, object-oriented programming language. It is popular because of its simple syntax, readability, large community support, extensive libraries, and use in web development, data science, AI, automation, and more.

2. Give two real-world use cases where Python is commonly used.

- 1) Web Development (Django, Flask)
- 2) Data Analysis and Machine Learning (Pandas, NumPy, TensorFlow)

3. How do you install Python on your computer and write your first script in VS Code?

Download Python from python.org, install it, install VS Code and the Python extension, create a .py file, write `print('Hello World')`, and run it.

4. Explain how Python code is executed from writing to output.

The source code is written in a .py file, interpreted by the Python interpreter, converted to bytecode, executed by the Python Virtual Machine (PVM), and the output is displayed.

5. What are variables in Python, and how do you declare one?

Variables are names used to store data.

Example:

```
name='Richa'
```

```
age=22
```

6. List two rules you must follow when naming variables in Python.

- 1) Variable names cannot start with a number.
- 2) Variable names cannot contain spaces or special characters except underscore (_).

7. What is the difference between primitive and non-primitive data types in Python?

Primitive: int, float, bool, str. Non-primitive: list, tuple, set, dictionary.

8. Write Python code to create variables for a student profile containing name, age, and enrolled course.

```
name='Richa'
```

```
age=22
```

```
course='Python Backend'
```

9. What does the type() function return in Python? Give an example.

It returns the data type of a variable.

Example:

```
x=10
print(type(x))
```

10. How do you get input from a user in Python, and how do you display output?

Use input() for input and print() for output.

Example:

```
name=input('Enter name: ')
print(name)
```

11. How would you convert a string input to an integer in Python?

```
num=int(input('Enter number: '))
```

12. Write a Python program that takes two numbers from the user and prints their sum.

```
a=int(input('Enter first number: '))
b=int(input('Enter second number: '))
print('Sum =',a+b)
```

13. What is an escape character in Python strings? Give an example.

Escape characters start with backslash. Example: \n creates a new line.

14. Explain the difference between if, elif, and else statements in Python.

if checks the first condition, elif checks additional conditions, else executes when all conditions are False.

15. Write a Python program that checks if a number entered by the user is positive, negative, or zero.

```
num=int(input('Enter number: '))
if num>0:
    print('Positive')
elif num<0:
    print('Negative')
else:
    print('Zero')
```

16. Describe how you would use nested if statements. Give a simple example.

Nested if means an if inside another if.

Example:

```
age=20
```

```
if age>=18:
    if age<60:
        print('Adult')
```

17. Write a Python program to determine if someone is eligible to donate blood.

```
age=int(input('Age: '))
weight=int(input('Weight: '))
if age>=18 and weight>=50:
    print('Eligible')
else:
    print('Not Eligible')
```

18. What is a for loop in Python used for?

A for loop is used to iterate over sequences such as lists, tuples, strings, or ranges.

19. Write a for loop to print each fruit name from a list: ['apple', 'banana', 'cherry'].

```
fruits=['apple','banana','cherry']
for fruit in fruits:
    print(fruit)
```

20. How can you use a for loop to count the number of vowels in a string?

```
text=input('Enter string: ')
count=0
for ch in text.lower():
    if ch in 'aeiou':
        count+=1
print(count)
```

21. What is the difference between a for loop and a while loop in Python?

for loop is used when the number of iterations is known; while loop is used when iterations depend on a condition.

22. Write a while loop to print numbers from 1 to 5.

```
i=1
while i<=5:
    print(i)
    i+=1
```

23. How would you create a simple star pattern like this using a while loop? *

```
** ***
```

```
i=1
while i<=3:
    print('*'*i)
    i+=1
```

24. What happens if the condition in a while loop never becomes False?

The loop becomes an infinite loop and runs continuously.

25. Explain the use of break, continue, and pass statements in loops.

break exits the loop, continue skips the current iteration, pass does nothing and acts as a placeholder.

26. Write a for loop that prints each fruit in the list ['apple', 'banana', 'cherry'], but skips 'banana'.

```
fruits=['apple','banana','cherry']
for fruit in fruits:
    if fruit=='banana':
        continue
    print(fruit)
```

27. How would you use break to exit a loop when a specific value is found?

```
nums=[1,2,3,4]
for n in nums:
    if n==3:
        break
    print(n)
```

28. Give an example where pass would be useful in a Python loop or function.

```
def future_function():
    pass
```

29. How can you access the first and last character of a string in Python?

```
s='Python'
print(s[0])
print(s[-1])
```

30. Write Python code to reverse a string entered by the user.

```
s=input('Enter string: ')
print(s[::-1])
```

31. What is string slicing? Give an example to extract a substring.

String slicing extracts part of a string.

Example: `text='Python'; print(text[0:3])`

32. Name two string methods in Python and explain what they do.

`upper()` converts text to uppercase. `lower()` converts text to lowercase.

33. How do you create a list in Python and add a new item to it?

```
lst=[1,2,3]
```

```
lst.append(4)
```

34. Write code to remove the third element from a list.

```
lst=[10,20,30,40]
```

```
del lst[2]
```

35. Explain the difference between the `append()` and `insert()` methods for lists.

`append()` adds an item at the end. `insert(index,item)` adds an item at a specific position.

36. How would you sort a list of numbers in ascending order?

```
nums=[5,2,8,1]
```

```
nums.sort()
```

37. What is a tuple in Python, and how is it different from a list?

A tuple is an ordered collection that is immutable, while a list is mutable.

38. Write code to create a tuple with three elements and access its second value.

```
t=(10,20,30)
```

```
print(t[1])
```

39. How can you concatenate two tuples in Python?

```
t1=(1,2)
```

```
t2=(3,4)
```

```
print(t1+t2)
```

40. What happens if you try to change an element of a tuple after creation?

Python raises a `TypeError` because tuples are immutable.

41. What is a dictionary in Python, and how do you add a new key-value pair?

A dictionary stores data in key-value pairs.

```
student={'name':'Richa'}
```

```
student['age']=22
```

42. Write code to update the value of an existing key in a dictionary.

```
student={'age':22}
student['age']=23
```

43. How can you get all the keys and all the values from a dictionary?

```
d={'a':1,'b':2}
print(d.keys())
print(d.values())
```

44. Explain the difference between keys(), values(), and items() methods in dictionaries.

keys() returns keys, values() returns values, items() returns key-value pairs.

45. How do you define a function in Python that takes two numbers and returns their product?

```
def multiply(a,b):
    return a*b
```

46. What is a lambda function? Write a lambda to square a number.

A lambda is an anonymous function.
square=lambda x:x*x

47. How would you use map() to double every number in a list?

```
nums=[1,2,3]
result=list(map(lambda x:x*2, nums))
```

48. Give an example of using filter() with a lambda to get only even numbers from a list.

```
nums=[1,2,3,4,5,6]
even=list(filter(lambda x:x%2==0, nums))
```

49. How would you explain the concept of return values in Python functions?

A return value is the result sent back from a function using the return keyword.

50. Write a function that takes a list of numbers and returns the maximum value.

```
def find_max(nums):
    return max(nums)
```

51. How would you use reduce() to calculate the sum of all elements in a list?

```
from functools import reduce
nums=[1,2,3,4]
total=reduce(lambda a,b:a+b, nums)
```

52. When would you use a function instead of repeating code in your program?

Use functions when the same logic is needed multiple times. It improves reusability and maintainability.

53. How could you use ChatGPT or GitHub Copilot to help you write or debug Python code, and what steps would you take to ensure the code it suggests is correct?

Use AI tools to generate code, explanations, and debugging suggestions. Verify by understanding the code, testing different cases, checking documentation, and reviewing outputs.

54. If you need to quickly generate example data or boilerplate code for your Python project, how might AI tools help, and what would you double-check before using their output?

AI can generate sample datasets and starter code. Double-check accuracy, security, efficiency, and project requirements.

55. Suppose you need to process a list of student records and extract names of students who scored above 80. Which combination of data structures and control flow statements from this module would you use, and why?

Use a list of dictionaries with a for loop and if condition because it allows efficient storage and filtering.

Example:

```
students=[{'name':'A','score':90},{'name':'B','score':70}]
for s in students:
    if s['score']>80:
        print(s['name'])
```

56. How would you design a Python function that takes a string, checks if it is a palindrome, and returns True or False? Mention which string methods and control flow you would use.

```
def is_palindrome(text):
    text=text.lower()
    return text==text[::-1]
```

57. Imagine you are building a student management system. Which data type would you use to store each student's profile (name, age, courses), and how would you access or update their information efficiently?

Use a dictionary.

```
student={'name':'Richa','age':22,'courses':['Python']}
```

```
print(student['name'])  
student['age']=23
```